

AKSHAY MOHAN GUPTA

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Education

Central Peel SS

Sep 2022 - Jun 2026

Advanced Placement (AP)

Brampton, Canada

- Average: **99.50%** (4.0/4.0 unweighted GPA)
- Relevant Courses: AP Statistics, AP Computer Science, AP Calculus BC, AP Physics 2, AP Chemistry, Computer Engineering.

Experience

Software Engineering Intern

Jun 2025 - Sep 2025

Royal Bank of Canada (RBC)

Toronto, ON

- Boosted internal system efficiency by **70%** with an **NLP chat bot** for answering queries and auto filling forms.
- Fine-tuned a **transformer-based embedding model** (BERT) to enhance chatbot understanding and accuracy.
- Interface built using **React** and **Flask**, with **CI/CD pipelines** using **Jenkins** for automated deployment and testing.
- Created **dynamic SQL queries** on IBMDB2 to automate the purging of sensitive data (**100,000+ entries purged**).
- Worked with: Python, TensorFlow, SQL, JavaScript, Java, Flask, React, Postman, Jenkins, DBVisualizer, and PEGA.
- Projected to save **210K hours** and **\$3.5M** annually 📈. Received **Return Offer**.

Software Engineering Instructor

Jun 2024 – Present

Peel District School Board

Peel Region, ON

- Led district-wide STEM initiatives and taught 60+ students in **programming** (Python, JavaScript, C++).
- Designed a development curriculum where students built real apps using React, Flask, TensorFlow, and AWS.
- Taught **Arduino (C++)**, circuits, and **embedded systems** (PCB basics) to create physical devices.
- Delivered project-based workshops on coding, circuit building, and hardware-software integration.

National Space Society (NSS) Space Settlement Competition - ML Lead 🚀

Jun 2023 – Present

Machine Learning & CAD Lead

Brampton, Canada

- Placed 3rd against **26,000 students worldwide** and invited to the International Space Development Conference.
- Developed an **ML model** trained on 5000+ exoplanets from NASA to identify optimal exoplanets for colonization.
- Created **drones** in Blender then simulated them in **Gazebo** for autonomous resource collecting and mining.
- Designed the settlement in Blender, conducted **Finite Element (FEA) & Structural Analysis** in **SimScale**.

Award Winning Projects

HRT Macroplace Challenge (103 Globally) 🚀 | Python, TensorFlow, C++, GNN, RL, Nvidia DREAMplace, Docker

- Placed **103 globally** in the Hudson River Trading (HRT) x Partcl Macroplacement challenge.
- **Beat Simulated Annealing by 23.7%** average proxy cost across all 17 IBM benchmarks (peak **+26.5% on ibm02**).
- Built a **GNN + RL pipeline** for VLSI macro placement on 200,000+ node circuits, encoding netlist data as heterogeneous tripartite graphs (macros, nets, ports) with **KNN spatial edges** for congestion-aware reasoning.
- Integrated AutoDMP on **NVIDIA DREAMPlace** as a post-processing legalization engine to resolve macro overlaps.
- **Eliminated O(N²) compute** by vectorizing HPWL on tensor cores and offloading spatial queries to KNN routines

Shopify Minis Hackathon 🚀 | Shop Minis SDK, React, TypeScript, Tailwind, Fal AI

- Built **ShopFit** as a Shop Mini that uses AI to recommend outfits based on user data, shopify products, and more.
- Utilized the **Shop Minis SDK** in order to collect user data, access product data, and get relevant products.
- Implemented **AI virtual try-on** (photo + video) and theme-based search layered on Shopify product data.
- Learned from a Shopify Engineering manager and a Shopify Senior Product Operations Manager.
- Presented to **Farhan Thawar (Shopify Head of Engineering)** who liked the project & vision.

BramFire Labs (4th @ Amazon x BramHacks) 🚀 | Python, YOLOv11, Typescript, Google Gemini API, Flask, Node.js

- Placed **4th out of 125 teams** for an AI-powered wildfire detection platform via satellite data + drone imagery.
- Built a real-time geospatial data pipeline and fine-tuned a **YOLOv11** model for early fire detection with high recall.
- Developed a full-stack dashboard for live inference, alerting, and heat-map visualization of fire-risk regions.
- Implemented a Twilio notifications system to automatically text/call subscribed users when a fire is detected.

Skills

Programming: Python, C++, Java, SQL, JavaScript, Linux, Docker, PostgreSQL, Google ADK, LangChain.

Libraries: PyTorch, Tensorflow, Flask, Pandas, Numpy, Matplotlib, Scikit-Learn, OpenCV, Hugging Face, React.

Competitions/Certificates: Waterloo Math Contests (top 15%), DECA Internationals, Udemy Tensorflow, Harvard CS50

Fun Facts: I play on the 13th **best volleyball team in Ontario** (Peel Champions), play ice hockey, and enjoy investing!